HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Contributed to the design and implementation of hydrogeological conceptual models for environmental impact assessments across multiple client projects.
- Managed and analyzed environmental data, including water quality samples and water levels, using **Excel** and **basic databases** to track trends.
- Assisted senior hydrogeologists in conducting field investigations, including supervising borehole drilling and logging soil and rock samples.
- Organized field data and drafted initial summaries for technical reports and environmental assessments, supporting **hydrogeological impact assessments**.
- Developed **Python** scripts to automate data processing tasks and generate reports for environmental monitoring, reducing manual effort by approximately **40%**.
- Collaborated with cross-functional teams to support water resource management projects, gaining exposure to **groundwater modelling** and **hydro-geochemistry**.
- Containerized data analysis tools and scripts using **Docker** for consistent deployment across development and field teams.

SKILLS

Hydrogeology – Physical Hydrogeology, Contaminant Hydrogeology, Hydro-geochemistry, Groundwater Modelling, Hydrogeological Conceptual Modelling, Hydrogeological Impact Assessment

Data Analysis & Tools – Excel, Basic Databases, GIS mapping tools, Python, R, Pumping test analysis

Fieldwork & Monitoring – Groundwater and surface water monitoring, Water quality sampling, Water level measurement, Borehole drilling supervision, Soil and rock sample logging, Pumping tests

Reporting & Assessment – Technical report writing, Environmental Impact Assessment (EIA), Appropriate Assessment (AA), Groundwater quality assessment

Software Development – Python, R, Docker, Git, Shell scripting

PROJECTS

Groundwater Contamination Data Analysis Platform

- Developed a **Python**-based data analysis pipeline to process and visualize groundwater quality data from multiple field sites, integrating with **basic databases**.
- Automated the generation of trend analysis reports and spatial maps using **GIS mapping tools** and **Python libraries**, enabling quicker identification of contamination plumes.
- Implemented data validation checks within the pipeline to ensure accuracy and consistency of input samples, improving data integrity by an estimated **15%**.

Automated Borehole Logging and Geotechnical Data Aggregation

- Created a script using **Python** and **shell scripting** to aggregate borehole logs and geological data from various field input formats into a centralized repository.
- Designed a data structure for storing geological and geotechnical information, supporting future **borehole logging** and analysis requirements.
- Assisted in structuring data for **pumping test analysis** by organizing log sheets and field measurements into a consistent format.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)